

Instrumenting return

Week 4, Lecture 2 · LangoBee to \$10K MRR

Summer 2026 · Operating a Solo Language App from Zero Revenue

What we'll cover

- D1, D7, D30: precise definitions and why each interval matters
- Reading a cohort retention curve: flattening vs decaying
- Small-n honesty: what two cohort weeks can and cannot tell you
- LangoBee's real cohort table: May 18 and May 25 read together
- Setting up the D1 dashboard that will judge your hook

D1, D7, D30: the retention clock

Three intervals, three questions

Each interval answers a different question about your product:

- **D1 (day 1):** did the first experience create enough value to come back 24 hours later? This is the activation quality check.
- **D7 (day 7):** did the habit form in the first week? This is the early-retention check.
- **D30 (day 30):** is there a stable segment of users for whom this product is genuinely useful? This is the PMF check.

A product where D1 is 40%, D7 is 20%, and D30 is 15% is in a different place than one where D1 is 40%, D7 is 5%, and D30 is 1%. The second product has a first-week cliff. The first product is retaining a floor.

(Balfour, 2013)

What D1 measures at LangoBee

LangoBee's current June 2026 snapshot: approximately 24% of real signups ever return at all. That is not a D1 number; it is a lifetime return rate across cohorts.

D1 specifically asks: of users who had a qualifying session today, how many returned tomorrow?

This number is harder to read from aggregate cohort tables. It requires the PostHog retention chart with a "day" period. The D1 dashboard you will build in section is the first time LangoBee has had this number visible.

The consumer-transactional benchmark for D1: good products see 20-30%, great products see 30-50%. (Winters and Rachitsky, 2021)

The lifecycle model

Archana Madhavan's retention lifecycle framework (Amplitude, 2016) divides users into three distinct cohorts that aggregate curves collapse together:

- **New users:** first session in the measurement window. D1 is their most critical moment.
- **Current users:** users who returned in the prior period and returned again. CURR is this group.
- **Resurrected users:** users who went dormant and returned. Email reactivation targets this group.

The Quick Ratio (new plus resurrected divided by churned) is above 1.0 when the product grows despite churn. LangoBee's current Quick Ratio is below 1.0 because current users are not returning faster than they are leaving.

(Madhavan, 2016)

Reading the retention curve

Flattening vs decaying: the two shapes

A retention curve plots the percentage of a cohort that returned on each subsequent day or week after their first session.

Flattening curve: the percentage drops steeply from D0 to D7, then levels off. Some floor of users is retained indefinitely. This is the signature of product-market fit for a subset. The floor does not need to be high; even 10% floor retention means 10% of users found the product genuinely useful.

Decaying curve: the percentage continues dropping toward zero with no visible floor. The curve suggests no stable segment has formed. Every cohort will eventually churn completely.

(Balfour, 2013)

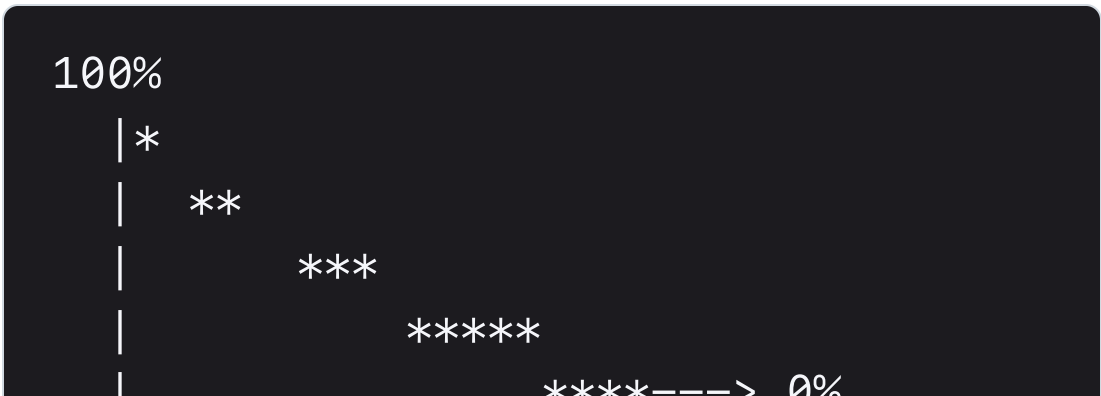
Shape comparison



Flattening curve

Indicates: a segment for whom this is habit-forming. Even a 10% floor is a signal.

Decaying curve



Small-n honesty

Two cohort weeks with under 15 activated users each cannot distinguish the two shapes. Here is why:

If May 18 returns 6/12 (50%) and May 25 returns 3/7 (43%), you have two points on a curve. Two points define a line, not a floor. The 95% confidence interval around a proportion of 6/12 at $n=12$ is roughly 21-79%. The two data points are statistically indistinguishable.

What you can say: the product is retaining some users across both cohorts. What you cannot say: whether there is a stable floor, whether the curve is improving, or whether activation quality is the driver.

The honest action: state what you observe, name the sample size, and continue collecting data.

(Balfour, 2013)

"The bottom line, with out retention, accelerating growth is meaningless.
Your retention curve is the best proof."

Brian Balfour (2013)

The Never Ending Road to Product Market Fit

LangoBee's real cohort table

The June 2026 snapshot from Supabase (de-alted, founder and bot accounts excluded):

Cohort week	Signups	Activated	Returned	Return rate
May 18	15	12	6	50% of activated
May 25	19	7	3	43% of activated
June 1	52*	1	0	0%

*June 1 signups are ~95% bot/spam. Real humans in that cohort: approximately 1. This is why de-alting matters: the raw signup count of 52 would make the June 1 cohort dominate any aggregate retention number.

The May cohorts are the only rows with signal. They are small.

What the cohort table says about activation

The June 1 cohort (52 signups, 1 activated) is the clearest possible illustration of why activation rate is a more honest leading indicator than signup count.

May 25 activated only 7 of 19 signups (37%). May 18 activated 12 of 15 (80%). The difference is not explained by the June bot wave; May 25 predates it.

Possible explanations: the onboarding flow changed between the two weeks; May 25 saw more paid-traffic signups with lower intent; the activation definition captures only a subset of genuine engagement.

This is a question for the section worksheet. The retention hook will not help users who do not activate.

Setting the D1 dashboard up to judge the hook

The D1 dashboard serves one purpose: measuring whether the hook you ship in the next sprint changes D1 return rate for activated users.

For the dashboard to be usable, it must:

- Exclude internal accounts (de-alting from week 1)
- Use a consistent qualifying event that does not change between now and your measurement date
- Show D1, D7, and D30 as fractions (not percentages alone) so small denominators are visible
- Be saved and pinned so you look at it every Monday

The worst outcome: you ship the hook, look at PostHog six weeks later, and find the dashboard was built with a different event definition than the one you tracked in the hook spec. Write the exact event names in both places before section.

Take-aways

1. D1 measures whether the first session created enough value to return tomorrow. D7 measures whether a weekly habit formed. D30 measures whether a stable segment exists.
2. A flattening retention curve, even at a low floor, signals product-market fit for some segment. A curve trending to zero means no stable segment has formed.
3. Two cohort weeks with n under 15 per cohort cannot distinguish flattening from decaying. State what you observe and name the sample size; do not over-conclude.
4. LangoBee's real data: May 18 at 50% return rate (6/12 activated), May 25 at 43% (3/7). The difference is within noise at these sample sizes.
5. The D1 dashboard must be built before you ship the hook, use the de-alted filter, and track the exact event names in the hook spec.

What's next

- **Section (this week):** Choose your hook, write its event spec, build the D1 dashboard
- **Reading:** Week 4 reading at </c/langobee-10k-mrr-26su/readings/wk04>
- **HW2 (in flight, due week 5):** Ship the chosen retention hook end to end
- **Week 5:** User conversations: talking to returners and one-and-dones